

Limits Problem Set A

COPY 1 OF 2

NAME

CLASS

DATE _____

Evaluate each limit. Show every algebraic step — an unsupported answer scores nothing.

Time 30 min · Total 26 marks

1 Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

3 marks

2 Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+16} - 4}{x}$

3 marks

3 Evaluate $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x^2 - 9}$

3 marks

4 Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 5x + 6}$

3 marks

5 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 2}{4x^2 + 7x - 1}$

3 marks

6 Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + 1}}{x + 3}$

3 marks

CHALLENGE SECTION

7 Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$ HARD

4 marks

8 Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$ HARD

4 marks

Time 30 min · Total 26 marks

3 marks

3 marks

3 marks

3 marks

5 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 5x + 2}{4x^2 + 7x - 1}$

3 marks

6 Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + 1}}{x + 3}$

3 marks

CHALLENGE SECTION

7 Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$ HARD

4 marks

8 Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$ HARD

4 marks